

|                   | Signal absent or degraded<br>Un gated<br>FP | Audit<br>FP | Reduction<br>(pp) |
|-------------------|---------------------------------------------|-------------|-------------------|
| No signal         | 100                                         | 0           | 100               |
| Weak signal       | 100                                         | 7           | 93                |
| Measurement error | 100                                         | 67          | 33                |

### Invalidity that preserves predictive performance

|                     |     |     |   |
|---------------------|-----|-----|---|
| Temporal drift      | 100 | 97  | 3 |
| Leakage             | 100 | 100 | 0 |
| Omitted confounding | 100 | 100 | 0 |
| Correlated features | 100 | 100 | 0 |
| Geographic shift    | 100 | 100 | 0 |

Values are percentages across 30 seeds.